EE2022 - Electrical Energy Systems

Lab Report

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5.1 AC Power Measurement for an R-L Load

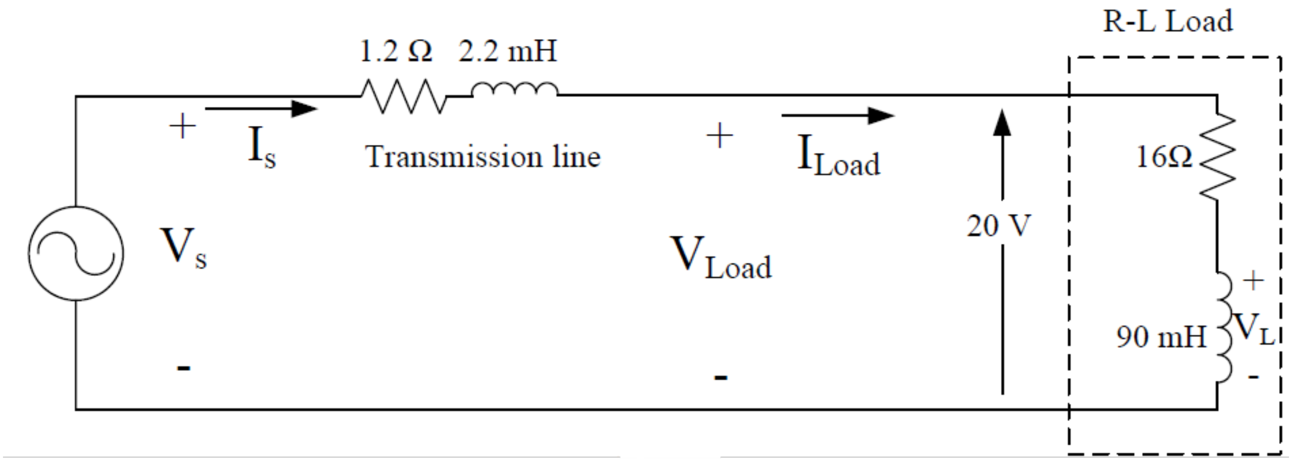


Figure 1: Original Circuit

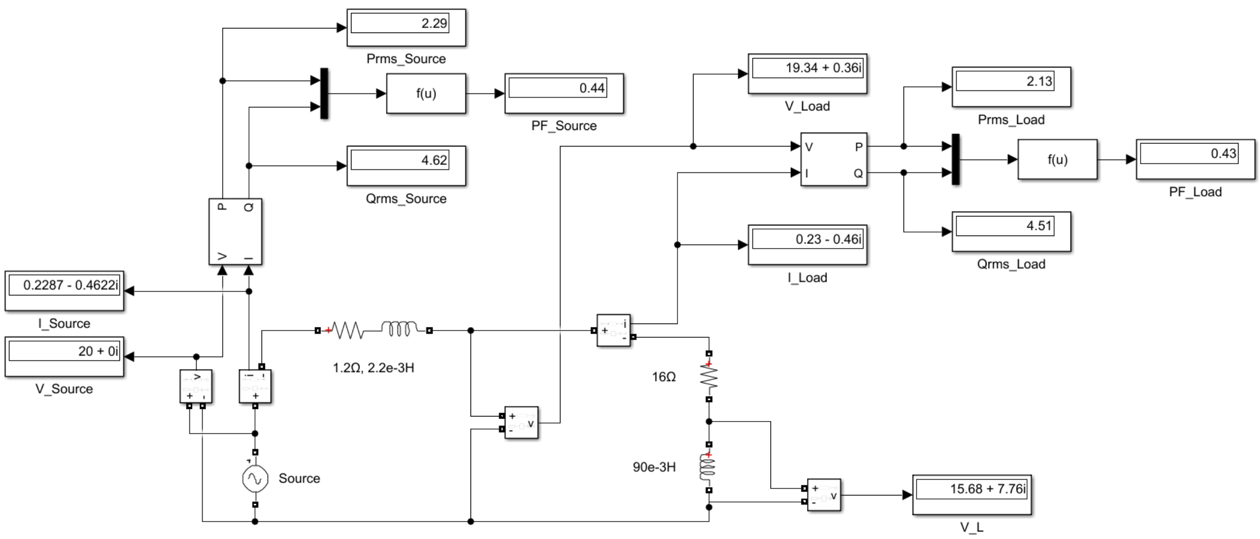


Figure 2: Simulink Simulation

Comparative Analysis Table

Parameter	Simulink Model	Experimental Results
V_{Source}	$V_S = 20 + j0 \implies \|V_S\| = 20\,\text{V}$.	$\|V_S\| = 20\,\text{V}$.
I_{Source}	$I_S = 0.2287 - j0.4622$ $\|I_S\| = \sqrt{0.2287^2 + (-0.4622)^2}$ $\|I_S\| = 0.5156\,\text{A}$	$\|I_S\| = 0.496\,\text{A}$.
P_{Source}	$P_{S,RMS} = 2.29\,\text{W}$. $P_S = 2.29 \times \sqrt{2} = 3.24\,\text{W}$.	$P_S = 5.8\,\text{W}$.
Q_{Source}	$Q_{S,RMS} = 4.62\,\text{var}$. $Q_S = 4.62 \times \sqrt{2} = 6.53\,\text{var}$.	$Q_S = 8.0\,\text{var}$.
Source PF	$\text{PF}_S = 0.44$.	$\text{PF}_S = 0.591$.

Parameter	Simulink Model	Experimental Results
V_{Load}	$V_L = 19.34 + j0.36$ V. $\lvert V_L\rvert = \sqrt{19.34^2 + 0.36^2}$ $\lvert V_L\rvert = 19.343\,\text{V}$	$\lvert V_L\rvert = 19.4\,\text{V}$.
I_{Load}	$\lvert I_L\rvert = \lvert I_S\rvert = 0.5156\,\text{A}$.	$\lvert I_L\rvert = 0.496\,\text{A}$.
P_{Load}	$P_{\text{Load,RMS}} = 2.13\,\text{W}$. $P_{\text{Load}} = 2.13 \times \sqrt{2} = 3.01\,\text{W}$.	$P_{\text{Load}} = 5.5\,\text{W}$.
Q_{Load}	$Q_{\text{Load,RMS}} = 4.51\,\text{var}$. $Q_{\text{Load}} = 4.51 \times \sqrt{2} = 6.38\,\text{var}$.	$Q_{\text{Load}} = 7.9\,\text{var}$.
Load PF	$\text{PF}_{\text{Load}} = 0.43$.	$\text{PF}_{\text{Load}} = 0.570$.
$V_{\text{L (across load)}}$	$V_L = 15.68 + j7.76$ V. $\lvert V_L\rvert = \sqrt{15.68^2 + 7.76^2}$ $\lvert V_L\rvert = 17.48\,\text{V}$	$\lvert V_L\rvert = 16.37\,\text{V}$.
I^2R Loss	Calculated as the difference between source and load real power. $P_{\text{loss}} = P_{\text{Source}} - P_{\text{Load}}$ $P_{\text{loss}} = 2.29 - 2.13 = 0.16\,\text{W}$.	Calculated from measured powers. $P_{\text{loss}} = P_{\text{Source}} - P_{\text{Load}}$ $P_{\text{loss}} = 5.8 - 5.5 = 0.3\,\text{W}$.
L_{Overall}	Calculated from total impedance. $Z_T = V_S/I_S$ $Z_T = 20/(0.2287 - j0.4622)$ $Z_T = 17.21 + j34.78\,\Omega$. $X_T = \omega L_T \implies 34.78 = (2\pi \cdot 60)L_T$. $L_T = 34.78/377 = 92.2\,\text{mH}$.	Calculated from total impedance. $\theta = \arccos(0.591) = 53.77°$. $Z_T = (20\angle 0°)/(0.496\angle -53.77°)$ $Z_T = 40.32\angle 53.77°\,\Omega$. $Z_T = 23.83 + j32.5\,\Omega$. $L_T = 32.5/377 = 86.2\,\text{mH}$.

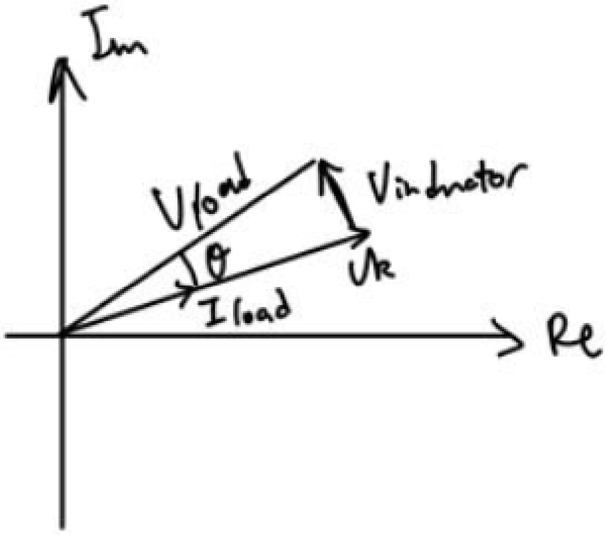


Figure 3: Phasor Diagram for R-L Load

We know the power factor of the load is 0.570, thus $\cos\theta = 0.570$ or $\theta = 55.25$

5.1.1 Component and Error Calculation

Based on the experimental measurements from Section 5.1 and a frequency of 60 Hz ($\omega \approx 377\,\mathrm{rad\cdot s^{-1}}$), we can calculate the component values and their errors.

Load Components

The R-L load values are calculated using the measured load power and current ($P_{\mathrm{Load}} = 5.5\,\mathrm{W}$, $Q_{\mathrm{Load}} = 7.9\,\mathrm{var}$, $I_{\mathrm{Load}} = 0.496\,\mathrm{A}$).

$$\begin{aligned}
R_{\mathrm{load,calc}} &= \frac{P_{\mathrm{Load}}}{I_{\mathrm{Load}}^2} = \frac{5.5\,\mathrm{W}}{(0.496\,\mathrm{A})^2} = 22.36\,\Omega \\
\mathrm{Error}_{R_{\mathrm{load}}} &= \left|\frac{22.36 - 16}{16}\right| \times 100\% = 39.75\% \\
X_{\mathrm{load,calc}} &= \frac{Q_{\mathrm{Load}}}{I_{\mathrm{Load}}^2} = \frac{7.9\,\mathrm{var}}{(0.496\,\mathrm{A})^2} = 32.11\,\Omega \\
L_{\mathrm{load,calc}} &= \frac{X_{\mathrm{load,calc}}}{\omega} = \frac{32.11\,\Omega}{377\,\mathrm{rad\cdot s^{-1}}} = 85.18\,\mathrm{mH} \\
\mathrm{Error}_{L_{\mathrm{load}}} &= \left|\frac{85.18 - 90}{90}\right| \times 100\% = 5.36\%
\end{aligned}$$

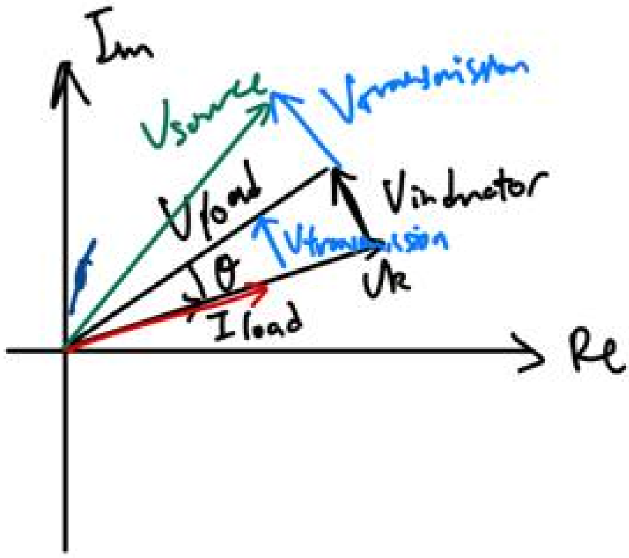


Figure 4: Phasor Diagram including Transmission Line

Transmission Line Components

The transmission line values are calculated from the power losses in the line, found by subtracting the load power from the source power ($I_S = 0.496\,\text{A}$).

$$P_{\text{line,loss}} = P_{\text{Source}} - P_{\text{Load}} = 5.8\,\text{W} - 5.5\,\text{W} = 0.3\,\text{W}$$
$$R_{\text{line,calc}} = \frac{P_{\text{line,loss}}}{I_S^2} = \frac{0.3\,\text{W}}{(0.496\,\text{A})^2} = 1.22\,\Omega$$
$$\text{Error}_{R_{\text{line}}} = \left|\frac{1.22 - 1.2}{1.2}\right| \times 100\% = 1.67\%$$
$$Q_{\text{line,loss}} = Q_{\text{Source}} - Q_{\text{Load}} = 8.0\,\text{var} - 7.9\,\text{var} = 0.1\,\text{var}$$
$$X_{\text{line,calc}} = \frac{Q_{\text{line,loss}}}{I_S^2} = \frac{0.1\,\text{var}}{(0.496\,\text{A})^2} = 0.406\,\Omega$$
$$L_{\text{line,calc}} = \frac{X_{\text{line,calc}}}{\omega} = \frac{0.406\,\Omega}{377\,\text{rad}\cdot\text{s}^{-1}} = 1.08\,\text{mH}$$
$$\text{Error}_{L_{\text{line}}} = \left|\frac{1.08 - 2.2}{2.2}\right| \times 100\% = 50.9\%$$

The experimental value for R_{line} is very accurate, while the load resistance and line inductance show significant error.

5.1.2 Power Triangle Analysis

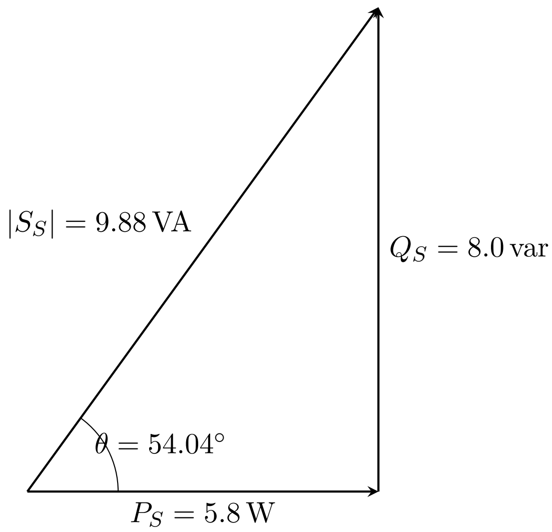
For the source in the R-L circuit, the experimental values are:

- **Real Power (P):** $P_S = 5.8\,\text{W}$
- **Reactive Power (Q):** $Q_S = 8.0\,\text{var}$
- **Power Factor (PF):** $\text{PF}_S = 0.591$ (lagging)

The Apparent Power (S) and the phasor angle (θ) are calculated as:

$$|S_S| = \sqrt{P_S^2 + Q_S^2} = \sqrt{5.8^2 + 8.0^2} = 9.88\,\text{VA}$$
$$\theta_{\text{pf}} = \arccos(\text{PF}) = \arccos(0.591) = 53.77^\circ$$
$$\theta_{\text{power}} = \arctan\left(\frac{Q_S}{P_S}\right) = \arctan\left(\frac{8.0}{5.8}\right) = 54.04^\circ$$

The angle θ represents the phase difference between the source voltage and current.



5.2 Power Factor Improvement

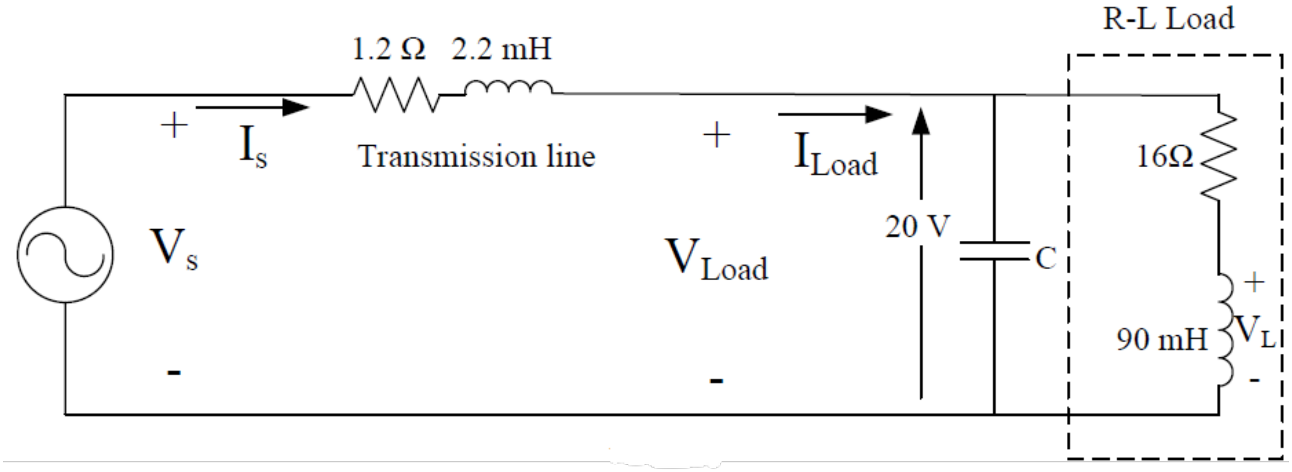


Figure 5: Original Circuit

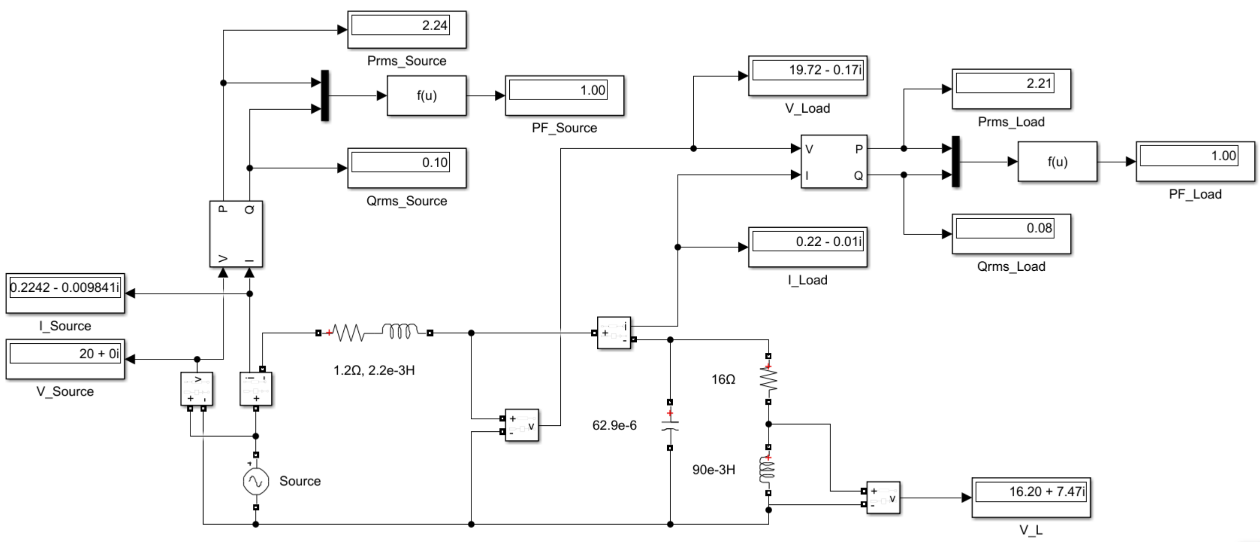


Figure 6: Simulink Simulation

To achieve a power factor of 1, the reactive power Q_L of the load must be cancelled by a capacitor. The required capacitance C is found using $C = \frac{Q_L}{(\omega |V_L|^2)}$. Based on the original load parameters from 5.1, the theoretically calculated capacitance was $C = \frac{7.9}{(\omega \times |20|^2)} = 62.9\,\mu\text{F}$. For the experiment, the closest available component was $C = 60\,\mu\text{F}$.

Comparative Analysis Table

Parameter	Simulink Model	Experimental Results
V_{Source}	$V_S = 20 + j0 \implies \lvert V_S\rvert = 20\,\text{V}$.	$\lvert V_S\rvert = 20.5\,\text{V}$.
I_{Source}	$I_S = 0.2242 - j0.009841$ $\lvert I_S\rvert = \sqrt{0.2242^2 + (-0.009841)^2}$ $\lvert I_S\rvert = 0.2244\,\text{A}$	$\lvert I_S\rvert = 0.328\,\text{A}$.
P_{Source}	$P_{S,RMS} = 2.24\,\text{W}$. $P_S = 2.24 \times \sqrt{2} = 3.17\,\text{W}$.	$P_S = 6.7\,\text{W}$.

Parameter	**Simulink Model**	**Experimental Results**
Q_{Source}	$Q_{S,RMS} = 0.10\,\text{var}$. $Q_S = 0.10 \times \sqrt{2} = 0.14\,\text{var}$.	$Q_S = 0\,\text{var}$.
Source PF	$\text{PF}_S = 1.00$.	$\text{PF}_S = 1.00$.
V_{Load}	$V_L = 19.72 - j0.17$ V. $\lvert V_L\rvert = \sqrt{19.72^2 + (-0.17)^2}$ $\lvert V_L\rvert = 19.72\,\text{V}$	$\lvert V_L\rvert = 20\,\text{V}$.
I_{Load}	$I_L = 0.22 - j0.01$ A. $\lvert I_L\rvert = \sqrt{0.22^2 + (-0.01)^2}$ $\lvert I_L\rvert = 0.2202\,\text{A}$	$\lvert I_L\rvert = 0.330\,\text{A}$.
P_{Load}	$P_{\text{Load,RMS}} = 2.21\,\text{W}$. $P_{\text{Load}} = 2.24 \times \sqrt{2} = 3.17\,\text{W}$.	$P_{\text{Load}} = 6.6\,\text{W}$.
Q_{Load}	$Q_{\text{Load}} = 0.08\,\text{var}$. $Q_{\text{Load}} = 0.08 \times \sqrt{2} = 0.11\,\text{var}$.	$Q_{\text{Load}} = 0\,\text{var}$.
Load PF	$\text{PF}_{\text{Load}} = 1.00$.	$\text{PF}_{\text{Load}} = 1.00$.
I^2R Loss	Calculated as the difference between source and load real power. $P_{\text{loss}} = P_{\text{Source}} - P_{\text{Load}}$ $P_{\text{loss}} = 2.24 - 2.21 = 0.03\,\text{W}$.	Calculated from measured powers. $P_{\text{loss}} = P_{\text{Source}} - P_{\text{Load}}$ $P_{\text{loss}} = 6.7 - 6.6 = 0.1\,\text{W}$.
C	$C = 62.9\,\mu\text{F}$. (Value set in Simulink model)	$C = 60\,\mu\text{F}$. (Closest available experimental component)

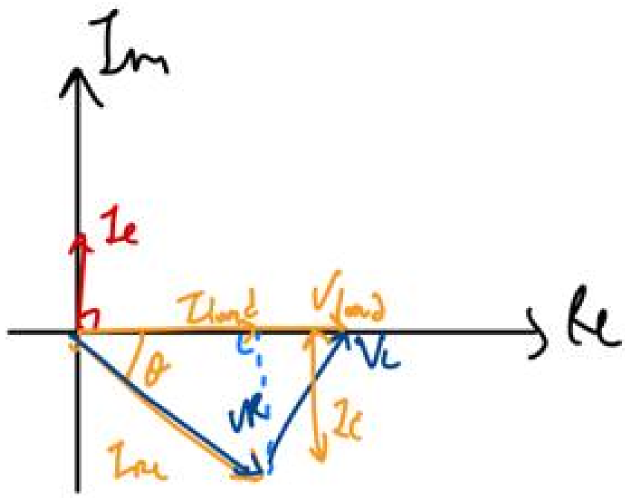


Figure 7: Phasor Diagram for Power Factor Correction

We know that $|I_C| = 2\pi f C |V_{load}| = 0.474\,\text{A}$. We can get the angle θ by using $\tan\theta = \frac{|I_c|}{I_{load}} = 1.436$ or $\theta = 55.15°$. Thus, we can get the resistance voltage by using $V_R = V_{load}\cos\theta = 11.43\,\text{V}$. From task 5.1, we get $R = 21.7\,\Omega$, thus the calculated I^2R is $\frac{V_R^2}{R} = \frac{11.49^2}{21.7} = 6.09 \approx 6.1\,\text{W}$.

5.2.1 Complex Power Analysis

Comparison of Complex Power (5.1 vs 5.2)

In Task 5.1, the circuit is heavily inductive. The source must supply both real power (P) and reactive power (Q) to the line and load.

- $S_{\text{Source}} = P_S + jQ_S = 5.8\,\text{W} + j8.0\,\text{var}$
- This results in a large apparent power $|S_S| = \sqrt{5.8^2 + 8.0^2} = 9.88\,\text{VA}$.
- The total source current I_S is high at 0.496 A to deliver this complex power.
- The power factor is poor at 0.591 (lagging).

In Task 5.2, a 60 µF capacitor is added in parallel to the load. This capacitor acts as a local source of reactive power, "generating" capacitive VARs that cancel the "consumed" inductive VARs from the load.

- $S_{\text{Source}} = P_S + jQ_S = 6.7\,\text{W} + j0\,\text{var}$
- The source now only needs to supply real power; the reactive power is handled locally. The experimental result of $Q_S = 0\,\text{var}$ and $PF = 1.00$ confirms this.
- The total apparent power $|S_S|$ drops to 6.7 VA, which is equal to the real power.
- Because the source no longer supplies reactive power, the total source current I_S is significantly reduced to 0.328 A.
- This current reduction directly decreases line losses:
  - $P_{\text{line,loss (5.1)}} = P_S - P_L = 5.8 - 5.5 = 0.3\,\text{W}$
  - $P_{\text{line,loss (5.2)}} = P_S - P_L = 6.7 - 6.6 = 0.1\,\text{W}$
- This reduction in loss (from 0.3 W to 0.1 W) improves the system's overall efficiency.

Effect of Larger Capacitance

The goal in Task 5.2 was to reach unity power factor ($PF = 1.0$), where the capacitor's reactive power Q_C perfectly cancels the load's reactive power Q_L.

$$Q_{\text{Total}} = Q_L - Q_C = 0$$

If a *larger capacitance* is used, Q_C will be greater than Q_L.

$$Q_{\text{Total}} = Q_L - Q_{C,\text{large}} < 0$$

The net complex power of the load $S_{\text{Load}} = P_{\text{Load}} - jQ_{\text{Total}}$ will become capacitive.

- The load power factor will decrease from 1.0, but it will become a leading power factor (e.g., 0.95 leading).
- The source will again have to supply reactive power, but this time it will be capacitive (leading Q).
- This re-introduces a reactive component to the source power S_S, increasing the apparent power $|S_S|$ from its minimum value.
- This, in turn, will increase the total source current I_S compared to the unity power factor case, leading to higher I^2R losses in the transmission line again.

5.2.2 Power Triangle Analysis

For the source in the PF-corrected circuit, the experimental values are:

- **Real Power (P):** $P_S = 6.7\,\mathrm{W}$
- **Reactive Power (Q):** $Q_S = 0\,\mathrm{var}$
- **Apparent Power (S):** $|S_S| = 6.7\,\mathrm{VA}$
- **Angle (θ):** $\theta = \arccos(P/|S|) = \arccos(6.7/6.7) = 0°$

This is a "degenerated" power triangle, as the reactive component is zero. The apparent power is equal to the real power, and the angle is 0°.

$$\xrightarrow[P_S = 6.7\,\mathrm{W}]{|S_S| = 6.7\,\mathrm{VA}}$$

$(Q_S = 0\,\mathrm{var},\ \theta = 0°)$

5.3a - Transformer 1 Short Circuit Test

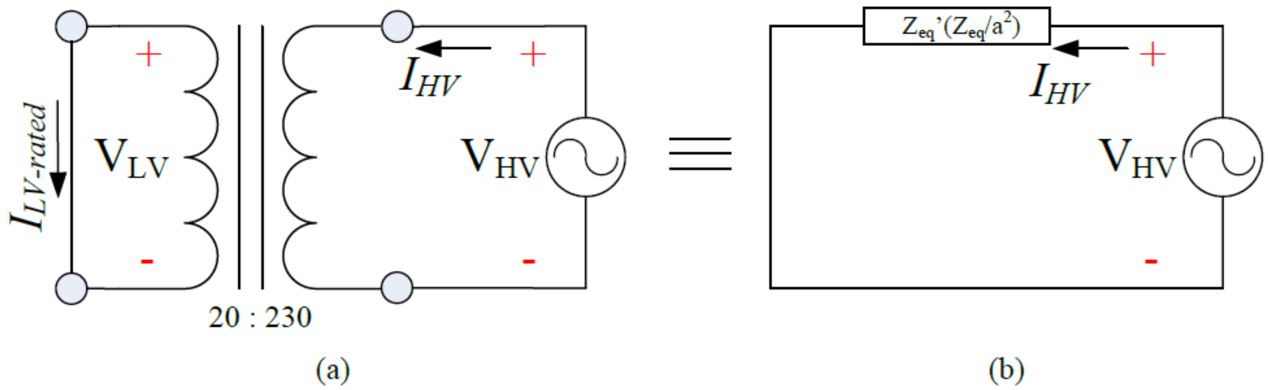


Figure 8: Transformer 1 Short Circuit

I_{LV} (A)	V_{LV} (V)	I_{HV} (A)	V_{HV} (V)	P_{HV} (W)	Q_{HV} (var)	P.F.$_{HV}$	R_{eq} (Ω)	X_{eq} (Ω)
3.002	0	0.250	12.4	3.2	0	1	51.2	0

Calculations for Transformer 1 SC Test:

$$Z'_{eq} = \frac{V_{HV}}{I_{HV}} = \frac{12.4}{0.250} = 49.6\,\Omega$$
$$R_{eq} = \frac{P_{HV}}{I_{HV}^2} = \frac{3.2}{0.250^2} = 51.2\,\Omega$$
$$X_{eq} = \sqrt{Z'^2_{eq} - R^2_{eq}} = \sqrt{49.6^2 - 51.2^2} \approx 0\,\Omega \text{ (PF = 1)}$$

5.3b - Transformer 1 Open Circuit Test

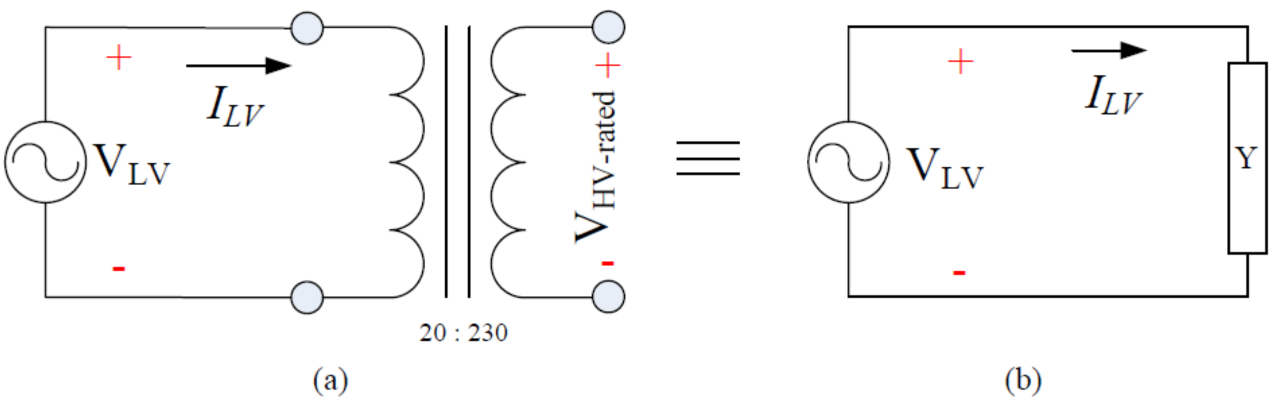


Figure 9: Transformer 1 Open Circuit

V_{HV} (V)	V_{LV} (V)	I_{LV} (A)	P_{LV} (W)	Q_{LV} (var)	P.F.$_{LV}$	R_m (Ω)	X_m (Ω)
230.6	19.5	0.660	4.1	12.2	0.317	92.7	31.2

Calculations for Transformer 1 OC Test:

$$Y = \frac{I_{LV}}{V_{LV}} = \frac{0.660}{19.5} = 0.0338\,\mathrm{S}$$
$$R_m = \frac{V_{LV}^2}{P_{LV}} = \frac{19.5^2}{4.1} = 92.7\,\Omega$$
$$X_m = \frac{V_{LV}^2}{Q_{LV}} = \frac{19.5^2}{12.2} = 31.2\,\Omega$$

5.3c - Transformer 2 Short Circuit Test

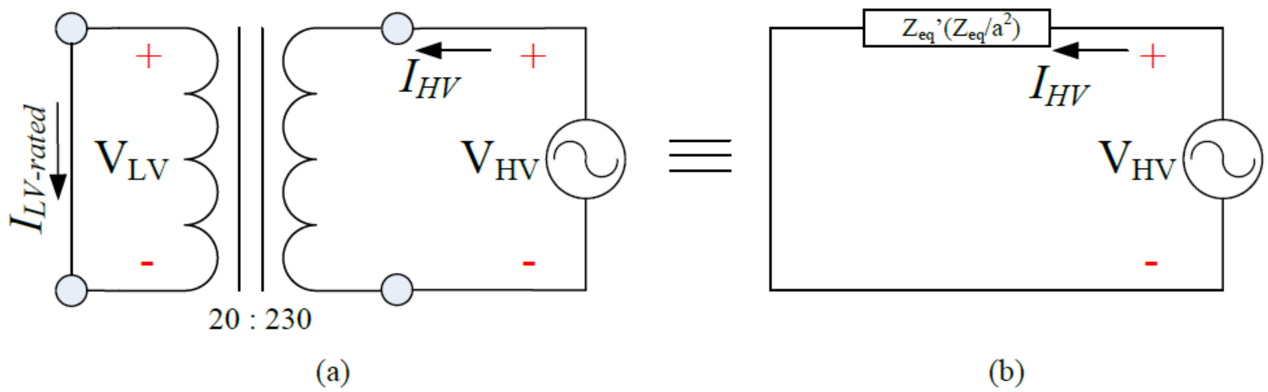


Figure 10: Transformer 2 Short Circuit

I_{LV} (A)	V_{LV} (V)	I_{HV} (A)	V_{HV} (V)	P_{HV} (W)	Q_{HV} (var)	P.F.$_{HV}$	R_{eq} (Ω)	X_{eq} (Ω)
3.015	0	0.250	12.5	3.2	0	1	51.2	0

Calculations for Transformer 2 SC Test:

$$Z'_{eq} = \frac{V_{HV}}{I_{HV}} = \frac{12.5}{0.250} = 50.0\,\Omega$$
$$R_{eq} = \frac{P_{HV}}{I_{HV}^2} = \frac{3.2}{0.250^2} = 51.2\,\Omega$$
$$X_{eq} = \sqrt{Z'^2_{eq} - R_{eq}^2} = \sqrt{50.0^2 - 51.2^2} \approx 0\,\Omega\ (\mathrm{PF} = 1)$$

5.3d - Transformer 2 Open Circuit Test

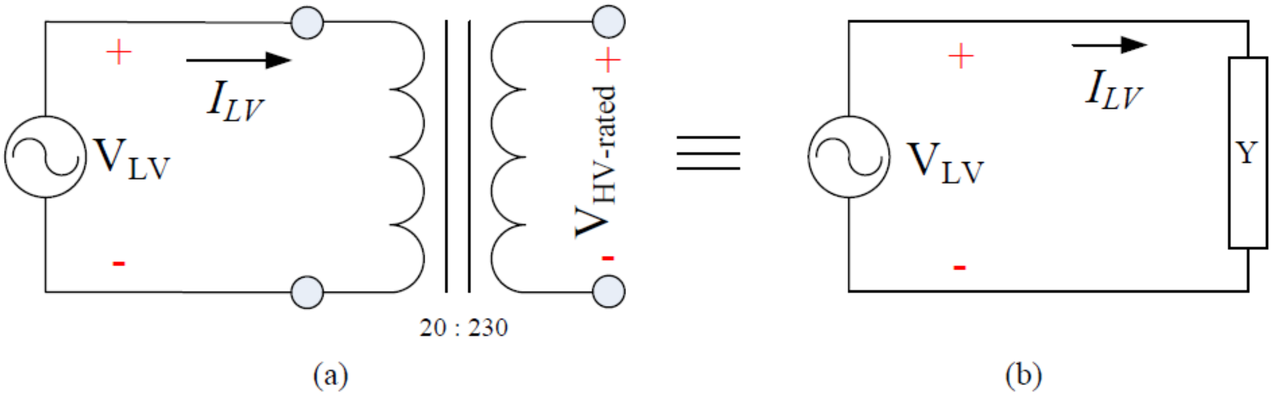


Figure 11: Transformer 2 Open Circuit

V_{HV} (V)	V_{LV} (V)	I_{LV} (A)	P_{LV} (W)	Q_{LV} (var)	P.F.$_{LV}$	R_m (Ω)	X_m (Ω)
230.7	19.5	0.649	4.1	11.9	0.322	92.7	32.0

Calculations for Transformer 2 OC Test:

$$Y = \frac{I_{LV}}{V_{LV}} = \frac{0.649}{19.5} = 0.0333\,\mathrm{S}$$

$$R_m = \frac{V_{LV}^2}{P_{LV}} = \frac{19.5^2}{4.1} = 92.7\,\Omega$$

$$X_m = \frac{V_{LV}^2}{Q_{LV}} = \frac{19.5^2}{11.9} = 32.0\,\Omega$$

5.4a Estimated Loss in Transmission Line Without Transformer

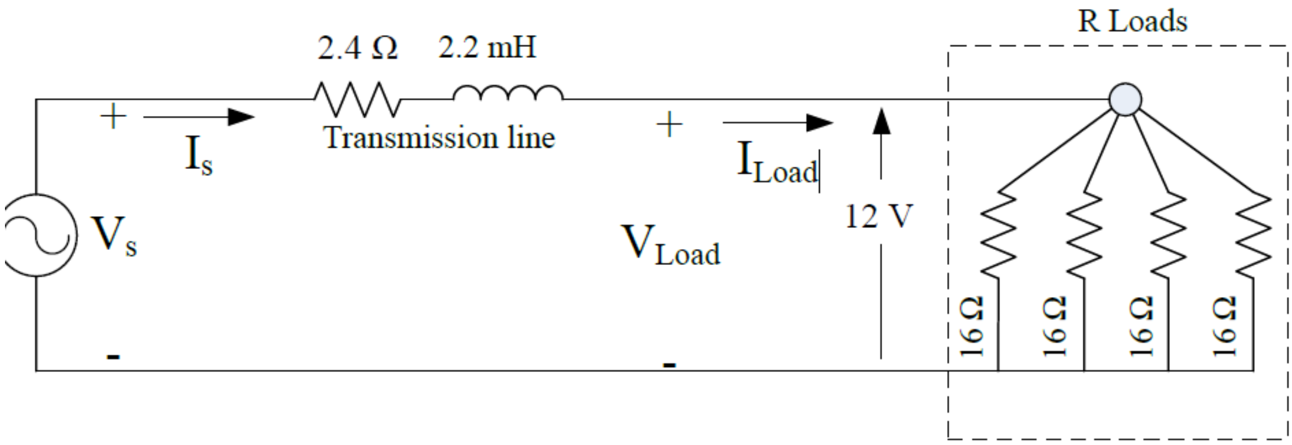


Figure 12: Without Transformer Circuit

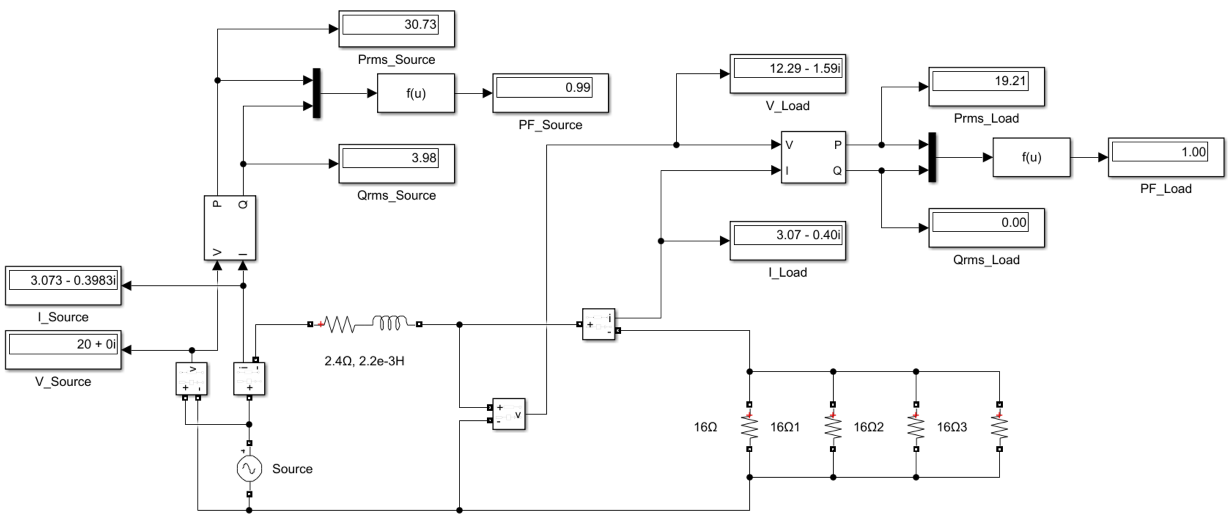


Figure 13: Without Transformer Simulation

Comparative Analysis Table

Parameter	Simulink Model	Experimental Results
V_{Source}	$V_S = 20 + j0 \implies \lvert V_S \rvert = 20.0\,\text{V}$.	$\lvert V_S \rvert = 19.7\,\text{V}$.
I_{Source}	$I_S = 3.073 - j0.3983$ $\lvert I_S \rvert = \sqrt{3.073^2 + (-0.3983)^2}$ $\lvert I_S \rvert = 3.099\,\text{A}$	$\lvert I_S \rvert = 3.038\,\text{A}$.
P_{Source}	$P_{S,RMS} = 30.73\,\text{W}$ $P_S = 30.73 \times \sqrt{2} = 43.46\,\text{W}$.	$P_S = 60.1\,\text{W}$.
Q_{Source}	$Q_{S,RMS} = 3.98\,\text{var}$ $Q_S = 3.98 \times \sqrt{2} = 7.96\,\text{var}$.	$Q_S = 0.0\,\text{var}$.
Source PF	$\text{PF}_S = 0.99$.	$\text{PF}_S = 1.0$.

Parameter	Simulink Model	Experimental Results
I_{Line}	$I_L = 3.07 - j0.40$ $\lvert I_L\rvert = \sqrt{3.07^2 + (-0.40)^2}$ $\lvert I_L\rvert = 3.096\,\text{A}$	$\lvert I_{\text{Line}}\rvert = 3.036\,\text{A}$.
V_{Load}	$V_L = 12.29 - j1.59$ $\lvert V_L\rvert = \sqrt{12.29^2 + (-1.59)^2}$ $\lvert V_L\rvert = 12.39\,\text{V}$	$\lvert V_L\rvert = 12.0\,\text{V}$.
I_{Load}	$\lvert I_L\rvert = 3.096\,\text{A}$	$\lvert I_L\rvert = 3.026\,\text{A}$.
P_{Load}	$P_{L,RMS} = 19.21\,\text{W}$. $P_L = 19.21 \times \sqrt{2} = 27.17\,\text{W}$.	$P_L = 36.4\,\text{W}$.
I^2R_{Line} Loss	Calculated from power difference: $P_{\text{loss}} = P_{\text{Source}} - P_{\text{Load}}$ $P_{\text{loss}} = 61.46 - 38.42 = 23.04\,\text{W}$.	Calculated from power difference: $P_{\text{loss}} = P_{\text{Source}} - P_{\text{Load}}$ $P_{\text{loss}} = 60.1 - 36.4 = 23.7\,\text{W}$.

Justification for Differences

1. **High Current in 5.4.a:** Without Transformers, the entire power must be transmitted at a low voltage (19.7 V). To deliver 60.1 W, a high current ($I_S = 3.038\,\text{A}$) is required.

2. **High I^2R Losses:** Power loss in the transmission line is calculated as $P_{\text{loss}} = I_{\text{line}}^2 R_{\text{line}}$. Because the current (3.038 A) is high, the power loss is substantial (23.7 W), representing a large portion of the total power.

3. **High Voltage Drop:** This high current also causes a large voltage drop across the line's impedance ($V_{\text{drop}} = I_{\text{line}} \cdot Z_{\text{line}}$). This is seen in the 7.7 V drop (39.08 %), which is highly inefficient.

5.4b Estimating Loss in Transmission Line with Transformer

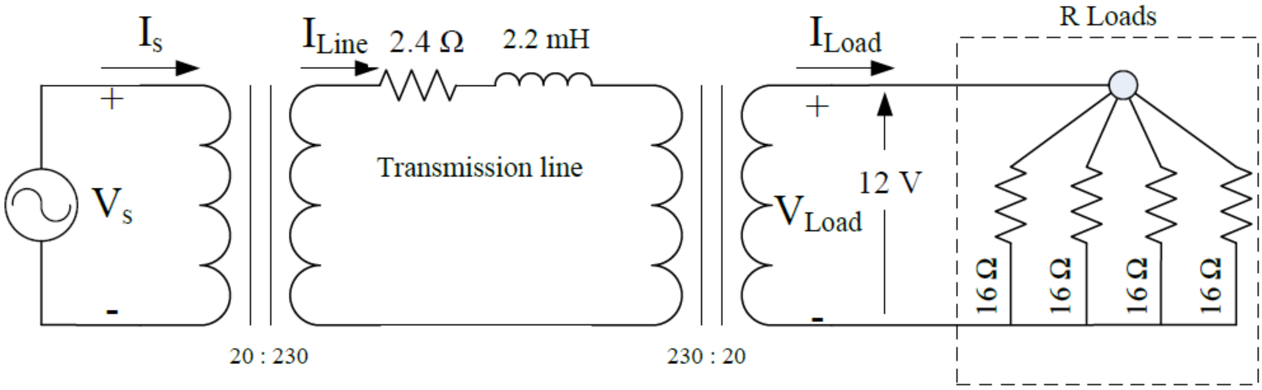


Figure 14: With Transformer Circuit

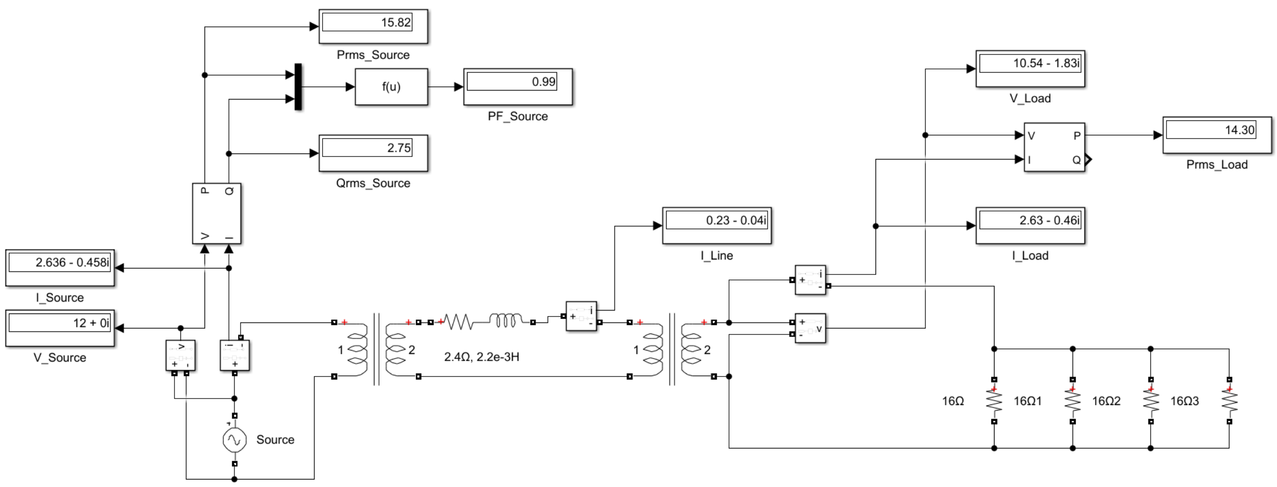


Figure 15: With Transformer Simulation

Comparative Analysis Table (With Transformers)

Parameter	**Simulink Model**	**Experimental Results**
V_{Source}	$V_S = 12 + j0 \implies \|V_S\| = 12.0\,\text{V}.$	$\|V_S\| = 14.5\,\text{V}.$
I_{Source}	$I_S = 2.636 - j0.458$ $\|I_S\| = \sqrt{2.636^2 + (-0.458)^2}$ $\|I_S\| = 2.676\,\text{A}$	$\|I_S\| = 3.318\,\text{A}.$
P_{Source}	$P_{S,RMS} = 15.82\,\text{W}.$ $P_S = 15.82 \times \sqrt{2} = 22.37\,\text{W}.$	$P_S = 48.0\,\text{W}.$
Q_{Source}	$Q_{S,RMS} = 2.75\,\text{var}.$ $Q_S = 2.75 \times \sqrt{2} = 3.89\,\text{var}.$	$Q_S = 2.2\,\text{var}.$
Source PF	$\text{PF}_S = 0.99.$	$\text{PF}_S = 0.999.$
I_{Line}	$I_{\text{Line}} = 0.23 - j0.04$ $\|I_{\text{Line}}\| = \sqrt{0.23^2 + (-0.04)^2}$ $\|I_{\text{Line}}\| = 0.233\,\text{A}$	$\|I_{\text{Line}}\| = 0.263\,\text{A}.$

Parameter	Simulink Model	Experimental Results
V_{Load}	$V_L = 10.54 - j1.83$ $\lvert V_L \rvert = \sqrt{10.54^2 + (-1.83)^2}$ $\lvert V_L \rvert = 10.70\,\text{V}$	$\lvert V_L \rvert = 12.0\,\text{V}$.
I_{Load}	$I_L = 2.63 - j0.46$ $\lvert I_L \rvert = \sqrt{2.63^2 + (-0.46)^2}$ $\lvert I_L \rvert = 2.670\,\text{A}$	$\lvert I_L \rvert = 3.011\,\text{A}$.
P_{Load}	$P_{L,RMS} = 14.30\,\text{W}$. $P_L = 14.30 \times \sqrt{2} = 20.22\,\text{W}$.	$P_L = 36.3\,\text{W}$.
$I^2 R_{\text{Line}}$ Loss	Calculated from line current: $P_{\text{line}} = I^2_{\text{Line}} \cdot R_{\text{line}}$ $P_{\text{line}} = (0.233\,\text{A})^2 \cdot 2.4\,\Omega = 0.130\,\text{W}$.	Calculated from line current: $P_{\text{line}} = I^2_{\text{Line}} \cdot R_{\text{line}}$ $P_{\text{line}} = (0.263\,\text{A})^2 \cdot 2.4\,\Omega = 0.166\,\text{W}$.
$P_{X'\text{mer}}$ Loss	Calculated from power difference: $P_{\text{total loss}} = P_S - P_L = 15.82 - 14.30 = 1.52\,\text{W}$. $P_{\text{tx}} = P_{\text{total loss}} - P_{\text{line}} = 1.52 - 0.130 = 1.39\,\text{W}$.	Calculated from power difference: $P_{\text{total loss}} = P_S - P_L = 48.0 - 36.3 = 11.7\,\text{W}$. $P_{\text{tx}} = P_{\text{total loss}} - P_{\text{line}} = 11.7 - 0.166 = 11.534\,\text{W}$.

Comparative Analysis

(i) Total Real Power Losses in Transmission Line

The real power loss in the transmission line is calculated using $P_{\text{line}} = I^2_{\text{Line}} \cdot R_{\text{line}}$, where $R_{\text{line}} = 2.4\,\Omega$.

- **Section 5.4.a (Without Transformers):**

$$P_{\text{line,a}} = (I_{\text{Line,a}})^2 \cdot R_{\text{line}} = (3.036\,\text{A})^2 \cdot 2.4\,\Omega = 9.217\,\text{A}^2 \cdot 2.4\,\Omega = 22.12\,\text{W}$$

- **Section 5.4.b (With Transformers):**

$$P_{\text{line,b}} = (I_{\text{Line,b}})^2 \cdot R_{\text{line}} = (0.263\,\text{A})^2 \cdot 2.4\,\Omega = 0.069\,\text{A}^2 \cdot 2.4\,\Omega = 0.166\,\text{W}$$

Comparison: The power loss in the line is dramatically reduced from 22.12 W to only 0.166 W by using transformers.

(ii) Real Power Loss Due to Transformer

The transformer loss is the total system loss ($P_S - P_L$) minus the transmission line loss (P_{line}).

- **Section 5.4.a (Without Transformers):**

$$P_{\text{tx,a}} = 0\,\text{W (No transformers are used.)}$$

- **Section 5.4.b (With Transformers):**

$$\begin{aligned} P_{\text{total loss,b}} &= P_{\text{Source,b}} - P_{\text{Load,b}} = 48.0\,\text{W} - 36.3\,\text{W} = 11.7\,\text{W} \\ P_{\text{tx,b}} &= P_{\text{total loss,b}} - P_{\text{line,b}} = 11.7\,\text{W} - 0.166\,\text{W} = 11.534\,\text{W} \end{aligned}$$

Quantification: The two transformers introduce a combined 11.53 W of real power loss.

(iii) Percentage of Efficiency

Efficiency (η) is the ratio of power received by the load to the power sent by the source ($\eta = P_L/P_S$).

- **Section 5.4.a (Without Transformers):**

$$\eta_a = \frac{P_{\text{Load,a}}}{P_{\text{Source,a}}} = \frac{36.4\,\text{W}}{60.1\,\text{W}} \approx 0.6057 \implies \mathbf{60.57}\%$$

- **Section 5.4.b (With Transformers):**

$$\eta_b = \frac{P_{\text{Load,b}}}{P_{\text{Source,b}}} = \frac{36.3\,\text{W}}{48.0\,\text{W}} \approx 0.7563 \implies \mathbf{75.63}\%$$

Comparison: The overall system efficiency is much higher with transformers (75.63 %) than without (60.57 %).

(iv) Percentage of Voltage Drop (Voltage Regulation)

- **Section 5.4.a (Without Transformers):** The voltage drop is the difference between source and load voltage.

$$\begin{aligned} V_{\text{drop,a}} &= V_{\text{Source,a}} - V_{\text{Load,a}} = 19.7\,\text{V} - 12.0\,\text{V} = 7.7\,\text{V} \\ \%V_{\text{drop,a}} &= \left(\frac{V_{\text{drop,a}}}{V_{\text{Source,a}}}\right) \times 100\% = \left(\frac{7.7\,\text{V}}{19.7\,\text{V}}\right) \times 100\% = \mathbf{39.08}\% \end{aligned}$$

- **Section 5.4.b (With Transformers):** Here, we calculate the voltage drop on the transmission line itself and compare it to the stepped-up voltage.

$$Z_{\text{line}} = R + j\omega L = 2.4 + j(2\pi \cdot 60 \cdot 2.2 \times 10^{-3}\,\text{H}) = 2.4 + j0.829\,\Omega$$
$$|Z_{\text{line}}| = \sqrt{2.4^2 + 0.829^2} = 2.54\,\Omega$$
$$V_{\text{drop,line,b}} = I_{\text{Line,b}} \cdot |Z_{\text{line}}| = 0.263\,\text{A} \cdot 2.54\,\Omega = 0.668\,\text{V}$$

The "sending" voltage on the high-voltage line is the source voltage stepped up: $V_{\text{line,send}} \approx V_S \times (\text{turns ratio}) = 14.5\,\text{V} \times (230/20) = 166.75\,\text{V}$

$$\%V_{\text{drop,b}} = \left(\frac{V_{\text{drop,line,b}}}{V_{\text{line,send}}}\right) \times 100\% = \left(\frac{0.668\,\text{V}}{166.75\,\text{V}}\right) \times 100\% = \mathbf{0.40\%}$$

Justification for Differences:

1. **Case 5.4.a:** Power is transmitted at low voltage ($\approx 20\,\text{V}$). To deliver the load power, a high current ($I_{\text{Line}} = 3.036\,\text{A}$) is required. This high current causes massive power loss ($P = I^2R \implies 22.12\,\text{W}$) and a severe voltage drop ($39.08\,\%$) across the line.

2. **Case 5.4.b):** A step-up transformer increases the voltage to a high level ($\approx 167\,\text{V}$) for transmission. Since power is constant ($P = VI$), this high voltage requires a very low current ($I_{\text{Line}} = 0.263\,\text{A}$). This tiny current results in negligible power loss ($P = I^2R \implies 0.166\,\text{W}$) and an almost zero voltage drop ($0.40\,\%$) on the line.

7.7 Experimentation Error

Several possible sources of error could have influenced the experimental results:

1. **Component and Wiring Errors:** The use of long connecting wires in the laboratory setup, particularly along the transmission line, introduces additional and unaccounted resistance and inductance. These parasitic effects increase the total I^2R losses and voltage drop, resulting in slightly higher losses than those calculated using the component specifications alone.

    - **Improvement:** Use the shortest practical wire lengths with suitable gauge ratings to minimize parasitic resistance and inductance.

2. **Transformer Core Hysteresis:** Real transformer cores exhibit hysteresis, meaning some magnetic flux can remain in the core from a previous test. This residual magnetism can affect the magnetization current and core loss measurements, especially during the open-circuit test.

    - **Improvement:** Demagnetize the core before testing by gradually increasing the voltage to the rated value using an AC autotransformer, then slowly reducing it back to zero.

3. **Transformer Core Saturation:** If the applied voltage (in the open-circuit test) or current (in the short-circuit test) exceeds the rated value, the core may enter saturation. In this state, the core's magnetic characteristics become highly non-linear, causing excessive magnetization currents and inaccurate readings.

    - **Improvement:** Ensure that all tests are conducted strictly within the rated voltage and current limits of the transformer.

4. **Instrument Measurement Errors:** Measuring instruments such as multimeters and power meters have inherent accuracy limits. Even small deviations due to calibration drift or poor connections can introduce discrepancies in the recorded values.

    - **Improvement:** Use recently calibrated, high-precision instruments and ensure that all electrical connections are secure to reduce contact resistance errors.

7.8 Advantages of Power Factor Improvement and Transformers

Advantages of Power Factor Improvement

Improving the power factor (PF) of an electrical load, in this case by connecting a parallel capacitor to offset inductive effects, offers several key benefits:

- **Reduced Line Current:** The capacitor locally supplies the reactive power (Q) demanded by the load, reducing the reactive component that must be delivered from the source. For the same real power (P), the apparent power (S) and total current ($I = S/V$) drawn from the source decrease.

- **Higher System Efficiency:** Since transmission losses are proportional to I^2R, lowering the current directly reduces these losses, improving the overall efficiency of the system.

- **Improved Voltage Regulation:** A lower current flow minimizes the voltage drop along transmission lines ($V_{\text{drop}} = I \cdot Z_{\text{line}}$), resulting in a more stable and consistent load voltage.

- **Increased System Capacity:** Because electrical infrastructure (such as transformers and cables) is rated by apparent power (VA), reducing S allows more real power (P) to be delivered using the same equipment—effectively increasing system capacity without hardware upgrades.

Advantages of Transformers in Power Systems

- **Facilitating High-Voltage Transmission:** By stepping up the voltage for transmission and stepping it down for end use, transformers allow large amounts of power ($P = VI$) to be sent with minimal current, drastically improving transmission efficiency.
- **Reducing Power Losses:** As seen in the comparison between Sections 5.4(a) and 5.4(b), high-voltage, low-current transmission dramatically lowers I^2R losses (e.g., 22.12 W vs. 0.166 W), highlighting how crucial voltage transformation is to minimizing losses.
- **Maintaining Voltage Stability:** Lower current flow also minimizes voltage drops ($V_{\text{drop}} = I \cdot Z_{\text{line}}$), ensuring that the voltage delivered to the load remains stable (e.g., 39.08 % vs. 0.40 % voltage drop).
- **Providing Electrical Isolation:** Transformers offer galvanic isolation between primary and secondary circuits, enhancing safety by separating end users from high-voltage systems and helping to contain electrical faults.